Name: ______________________

TRANSFORMATIONS, (REFLECTION)

Date: ________________

LO: I can reflect shapes in given mirror lines and describe reflections.

Reflection

A **reflection** flips a shape over a **mirror line**. Each point and its image are the same perpendicular distance from the mirror line.

Common mirror lines: x-axis (y = 0), y-axis (x = 0), y = x, y = -x, or lines like x = 2.

Quick examples

	(3, 2) reflected in y-axis (-3, 2)	x changes sign
	(3, 2) reflected in x-axis (3, -2)	y changes sign
	(3, 2) reflected in y = x (2, 3)	swap x and y
	Reflection in y-axis changes the y-coordinate	it changes x
	Reflected shapes change size	size stays the same

Key idea

Every point is the same distance from the mirror line as its image, measured perpendicularly.

y-axis: (x, y) (-x, y) | x-axis: (x, y) (x, -y)

The mirror line is the perpendicular bisector of each point and its image

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – in y-axis

Reflect (4, 3) in the y-axis.

The y-axis is x = 0

x-coordinate changes sign: 4 -4

y stays the same: (-4, 3)

(-4, 3)

Reflecting in the y-axis negates the x-coordinate.

Example 2 – in x-axis

Reflect (-2, 5) in the x-axis.

The x-axis is y = 0

y-coordinate changes sign: 5 -5

x stays: (-2, -5)

(-2, -5)

Reflecting in the x-axis negates the y-coordinate.

Example 3 – in y = x

Reflect (1, 4) in the line y = x.

For y = x, swap the coordinates

x becomes y and y becomes x

(1, 4) (4, 1)

(4, 1)

Reflecting in y = x swaps the x and y coordinates.

Example 4 – in vertical line

Reflect (5, 3) in the line x = 2.

Distance from x = 2: 5 - 2 = 3 units right

Image is 3 units left of x = 2: 2 - 3 = -1

(-1, 3)

(-1, 3)

Count the perpendicular distance and go the same distance the other side.

YOUR TURN – PRACTICE (deliberate practice)

1. Reflection in y-axis

Reflect each point in the y-axis.

a) (3, 5)

b) (-2, 4)

c) (7, -1)

d) (-6, 0)

2. Reflection in x-axis

Reflect each point in the x-axis.

a) (4, 2)

b) (1, -3)

c) (-5, 7)

d) (0, -4)

3. Reflection in y = x

Reflect each point in the line y = x.

a) (2, 5)

b) (1, 7)

c) (-3, 4)

d) (6, -2)

4. Reflection in other lines

Reflect each point in the given line.

a) (5, 3) in x = 3

b) (1, 4) in y = 2

c) (7, 1) in x = 4

d) (3, 6) in y = 4

5. Describing reflections

Describe the single reflection that maps the first point to the second.

a) (3, 2) (-3, 2)

b) (4, 5) (4, -5)

c) (1, 6) (6, 1)

d) (7, 3) (1, 3)

e) (2, 7) (2, 3)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Reflecting in the y-axis changes the sign of the x-coordinate ☐

Correct answer: __________

Reason:

b) Reflection changes the size of the shape ☐

Correct answer: __________

Reason: __________________

c) Reflecting in y = x swaps the x and y coordinates ☐

Correct answer: __________

Reason: __________________

d) Reflecting in the x-axis changes the x-coordinate ☐

Correct answer: __________

Reason: __________________

e) The mirror line is equidistant from a point and its image ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A triangle has vertices **A(1, 2), B(4, 2), C(4, 5)**. It is reflected in the line **y = x**. Find the coordinates of the image and describe the relationship between the triangle and its image.

Expression: ______________________

Simplified: ______________________

b) Point P(a, b) is reflected in the line **y = -x**. The image is P'(-b, -a). Verify this rule for the point **(3, 5)** and prove it works for any point.

Expression: ______________________

Simplified: ______________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!